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Epidemiology of severe acute respiratory infections in South Korea: a nationwide surveillance study

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Abstract

Background Severe acute respiratory infections (SARIs) are a critical public health concern due to their substantial morbidity and mortality rates across age groups. The objective of this study was to investigate the epidemiological characteristics, pathogen distribution, and clinical outcomes of SARIs in South Korea by analyzing nationwide surveillance data.

Methods We conducted a prospective, multicenter surveillance study from January 2017 to January 2023 across sentinel hospitals nationwide, coordinated by the Korea Disease Control and Prevention Agency (KDCA). Respiratory specimens were collected and tested for a panel of viral and bacterial pathogens. Clinical data and outcomes were analyzed according to pathogen detection status.

Results A total of 47,857 hospitalized patients with SARIs were enrolled based on standardized criteria: fever (≥ 38 °C), cough, and symptom onset within 10 days prior to hospitalization. Pathogens were identified in 52.5% of enrolled patients. The group that tested positive for pathogens was significantly younger (median age of 4.0 years) than the group that tested negative (median age of 59.0 years). Viral pathogens were more prevalent in children, whereas bacterial pathogens, such as Streptococcus pneumoniae, were more prevalent in older adults. Seasonal trends were observed: influenza and respiratory syncytial virus (RSV) peaked in winter; rhinovirus circulated year-round; and SARS-CoV-2 showed irregular peaks. Patients with confirmed pathogens had higher rates of pneumonia and mortality, as well as longer hospital stays ($P < 0.001$), compared to patients without confirmed pathogens. Antivirals were more commonly used in the pathogen-positive group, while antibiotics were predominantly prescribed to patients without confirmed pathogens. Geographic and sex-specific variations in pathogen distribution were also identified.

Conclusions This study provides a thorough examination of the epidemiology of SARIs in South Korea over a six-year period. The findings reveal notable differences in pathogen prevalence and related clinical outcomes by age, season, and region. These results underscore the importance of sustained nationwide surveillance and the development of targeted, seasonally adaptive public health strategies to mitigate the impact of SARIs on vulnerable populations.

Keywords Coronavirus, Epidemiology, Public health, Respiratory tract infections, South korea

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Background

Respiratory infections remain a major global health challenge, affecting millions annually. These infections range from mild upper respiratory tract illnesses to severe acute lower respiratory tract infections, such as pneumonia and bronchiolitis, which are leading causes of hospitalization and death [1], particularly among young children [2] and older adults [3, 4]. The impact of these infections is more pronounced in countries with limited healthcare resources, where acute respiratory infections represent a significant burden. The recent coronavirus disease 2019 (COVID-19) pandemic caused by SARS-CoV-2 has further highlighted the seriousness of respiratory infections and the importance of focused attention on robust surveillance and management strategies [5].

The epidemiology of respiratory infections is influenced by factors such as geography, population demographics, and seasonal patterns. The prevalence of the related pathogens—such as respiratory syncytial virus (RSV) [6], rhinovirus [7], and influenza [8]—varies over time and location. In South Korea, where demographic and seasonal variations are pronounced, these infections represent a considerable burden; however, large-scale studies across all age groups remain limited. This gap highlights the need for continuous, comprehensive surveillance to guide effective prevention and management strategies.

The ongoing COVID-19 pandemic has further emphasized the importance of respiratory pathogen surveillance to address the evolving nature of severe acute respiratory infections (SARIs) [9]. Notably, the COVID-19 pandemic greatly changed global surveillance for respiratory infections. In South Korea, intensified surveillance efforts may have contributed to increased detection of SARI cases. However, disruptions in hospital routines and shifts in testing priorities might have affected data collection trends over time.

SARIs significantly impact patient outcomes and health care systems. To address the gaps in our understanding of SARI epidemiology, we conducted a comprehensive nationwide prospective surveillance study in Korea from 2017 to 2023. In this study, which includes data from over 47,000 patients, we aimed to comprehensively analyze pathogen prevalence, clinical characteristics, and treatment patterns across diverse patient demographics.

A critical aspect of this study is the comparative analysis of SARI cases with and without pathogen detection. By examining the clinical characteristics and outcomes of these two groups, we aimed to elucidate the nuanced differences that emerge when microbiologic confirmation is available. Recognizing that the pathogen-negative group may include both true negative cases and undiagnosed infections, we investigated how the presence or absence of pathogen identification influences clinical severity, treatment approaches, and overall patient management. Through this multi-year surveillance, we sought to generate evidence that can inform more targeted and effective public health strategies for managing SARIs in the Korean population.

Methods

Hospital-based surveillance network

Nationwide surveillance for SARIs was conducted in South Korea from January 2017 to January 2023. The surveillance was conducted in collaboration with selected sentinel hospitals across the country, and the network gradually expanded over the years. Data collection and monitoring were managed by the Korean Disease Control and Prevention Agency (KDCA).

The number of participating institutions and the length of enrollment are shown below: In 2017, one hospital initiated data collection in Week 35, followed by participation from 24 hospitals in Week 36. In 2018, 13 hospitals were active during weeks 1–18 and 34–52, with one additional hospital contributing data during Weeks 19–21. In 2019, 13 facilities participated in Weeks 1–21 and 37–52, while 15 hospitals contributed during Weeks 19–36. The monitoring network expanded significantly in 2020, with 40 hospitals participating in Weeks 20–28, 41 in Weeks 29–48, and 42 hospitals (Table S1) beginning in Week 49. This level of participation was maintained in 2021 and 2023. Hospitals were selected based on geographic distribution, patient volume, and their capacity to conduct microbiological testing. The network was progressively expanded to enhance representativeness and ensure the inclusion of both urban and rural regions.

All centers adhered to standardized protocols for patient enrollment, sample collection, and data management. Regular training sessions, monitoring activities, and site visits were conducted to ensure consistent compliance with program guidelines. Because the number of sentinel hospitals varied slightly over time, we acknowledge that this may have influenced case ascertainment and data representativeness. This limitation has been addressed in the Discussion.

Patient enrollment and specimen testing

Patients diagnosed with SARIs were eligible for enrollment if they met the following criteria: onset of symptoms within 10 days of hospital admission; documented history of fever ≥ 38 °C; presence of cough symptoms; and hospitalization requirement. In total, 47,857 patients were enrolled in the surveillance database. The final sample size for analysis is reported in the Results section. Data were collected on the prevalence of various pathogens, including viral and bacterial species. The viral pathogens identified included adenovirus, human bocavirus, parainfluenza virus, RSV, influenza, rhinovirus,

human coronavirus, human metapneumovirus, and SARS-CoV-2. Bacterial pathogens of interest included *Mycoplasma pneumoniae*, *Chlamydia pneumoniae*, *Streptococcus pneumoniae*, and *Legionella pneumophila*.

Specimens, including nasopharyngeal swabs, sputum, and bronchoalveolar lavage fluid (BALF), were collected upon admission prior to initiating any therapeutic measures to ensure accurate pathogen detection. Routine microbiological evaluations, including blood and sputum cultures and Gram staining, were performed on all enrolled patients. Viral testing using PCR or rapid antigen assays for influenza, respiratory syncytial virus (RSV), and SARS-CoV-2 was conducted according to the standardized surveillance protocol. However, the specific testing panels varied by hospital based on available resources and clinical indications. All microbiological testing served dual purposes: surveillance data collection and clinical diagnosis. This ensured both data quality and optimal patient care.

Polymerase chain reaction (PCR) was used to detect viral pathogens, whereas antigen tests were used to diagnose influenza and SARS-COV-2. Bacterial cultures were performed on blood, sputum, and BALF to identify pathogens, such as *Streptococcus pneumoniae* and *Mycoplasma pneumoniae*. Additionally, urinary antigen testing for *Legionella pneumophila* and *Streptococcus pneumoniae* was performed to complement the diagnostic results in relevant cases.

Ethical considerations

This study was conducted in accordance with the ethical standards of the Declaration of Helsinki. The Institutional Review Board of Chungnam National University Hospital (No. 2023-03-076) reviewed and approved the study and waived the requirement of informed consent. All patient data were anonymized and treated with strict confidentiality to protect the privacy of the participants. Identifiable information was removed before analysis to ensure compliance with privacy regulations.

Data management and statistical analysis

Patient demographics, clinical symptoms, laboratory results, and treatment outcomes were recorded by trained personnel at each site and uploaded to a central database managed by the KDCA. The database was regularly reviewed for completeness and accuracy.

Statistical analyses were performed using SPSS version 25.0 (IBM Corp., Armonk, NY, USA). Categorical variables were summarized as frequencies and percentages, and continuous variables were presented as medians with interquartile ranges due to non-normal distribution. Group comparisons for categorical variables were performed using Pearson's chi-squared test or Fisher's exact test, as appropriate. Comparisons for continuous variables were made using analysis of variance for normally distributed data or the Kruskal–Wallis test for non-normally distributed data, depending on the normality of the data distribution, as assessed by the Shapiro–Wilk test. A two-tailed p-value of less than 0.05 was considered statistically significant. Additionally, stratified analyses were conducted by age group (7 years and younger, 8–18 years, 19–64 years, and 65 years and over) and sex to account for potential demographic confounding factors in pathogen distribution and clinical outcomes.

Results

Patients' baseline characteristics and clinical outcomes

In total, 47,857 patients were included. Of the enrolled patients, 52.5% ($n = 25,132$) had an identified SARI pathogen, while 47.5% ($n = 22,725$) had no SARI pathogen detected. Table 1 shows the patients' baseline characteristics. The SARI pathogen-detected group was younger, had a lower proportion of males, and included a higher proportion of individuals living in the capital region than in the group without detected SARI pathogens. Regarding symptoms, the SARI pathogen-detected group had a high prevalence of fever, cough, sputum production, and diarrhea, whereas dyspnea and other symptoms were less common in this group. Regarding comorbidities, patients without a detected SARI pathogen generally exhibited a higher prevalence of underlying conditions compared to those in the pathogen-detected group.

Table 2 summarizes the treatments and clinical outcomes of the patients. Antiviral agents were more commonly administered in the pathogen-detected group, while antibiotics were more frequently used among patients without detected SARI pathogens. In terms of outcomes, the pathogen-detected group exhibited a lower incidence of pneumonia, reduced mortality, shorter overall hospital stays, and fewer intensive care unit (ICU) admissions compared to the pathogen-negative group. Notably, however, the length of ICU stay was longer among patients with detected SARI pathogens.

Pathogen spectrum and age-specific patterns

Figure 1A shows the distribution of SARI species, identifying several viral and bacterial pathogens in the population. Rhinovirus was the most common viral pathogen (5,653 cases, 11.8%), followed by RSV (4,723 cases, 9.9%), SARS-CoV-2 (4,606 cases, 9.6%), and influenza virus (3,252 cases, 6.8%). Other viruses detected included adenovirus (1,922 cases, 4.0%), parainfluenza virus (1,903 cases, 4.0%), human metapneumovirus (1,545 cases, 3.2%), human bocavirus (1,430 cases, 3.0%), and human coronavirus (1,078 cases, 2.3%). *Mycoplasma pneumoniae* was the most common bacterial pathogen (3,765 cases, 7.9%), followed by *Streptococcus pneumoniae*

Table 1 Baseline characteristics of patient

Variables	All patients	SARI pathogen detected	No SARI pathogen detected	*P*-value
Number of patients	47,857	25,132 (52.5)	22,725 (47.5)	
Age (years)	18.0 (2.0–69.0)	4.0 (1.0–52.0)	59.0 (6.0–75.0)	< 0.001
Age group				
Children (≤ 7)	21,190	15,108 (60.1)	6,082 (26.8)	< 0.001
School-aged children (8–18)	2,739	1,597 (6.4)	1,142 (5.0)	< 0.001
Adults (19–64)	9,563	3,823 (15.2)	5,740 (25.3)	< 0.001
Older (≥ 65)	14,365	4,604 (18.3)	9,761 (43.0)	< 0.001
Male	27,553 (57.6)	14,006 (55.7)	13,547 (59.6)	< 0.001
Capital region[†]	27,223 (56.9)	14,525 (57.8)	12,698 (55.9)	< 0.001
Symptoms				
Fever	46,850 (97.9)	24,763 (98.5)	22,087 (97.2)	< 0.001
Cough	46,530 (97.2)	24,600 (97.9)	21,930 (96.5)	< 0.001
Sputum	26,106 (54.6)	14,403 (57.3)	11,703 (51.5)	< 0.001
Dyspnea	9,381 (19.6)	3,834 (15.3)	5,547 (24.4)	< 0.001
Myalgia	1,953 (4.1)	999 (4.0)	954 (4.2)	0.218
Diarrhea	2,581 (5.4)	1,426 (5.7)	1,155 (5.1)	0.004
Other symptoms	11,390 (5.4)	5,719 (22.8)	5,671 (25.0)	< 0.001
Comorbidities				
Hypertension	4,305 (9.0)	1,393 (5.5)	2,912 (12.8)	< 0.001
Diabetes mellitus	5,930 (12.4)	2,073 (8.2)	3,857 (17.0)	< 0.001
Chronic heart disease	3,971 (8.3)	1,497 (6.0)	2,474 (10.9)	< 0.001
Chronic lung disease	3,689 (7.7)	1,249 (5.0)	2,440 (10.7)	< 0.001
Chronic liver disease	811 (1.7)	284 (1.1)	527 (2.3)	< 0.001
Chronic neurologic disease	3,945 (8.2)	1,453 (5.8)	2,492 (11.0)	< 0.001
Chronic kidney disease	2,046 (4.3)	796 (3.2)	1,250 (5.5)	< 0.001
Solid organ tumor	3,118 (6.5)	722 (2.9)	2,396 (10.5)	< 0.001
Hematologic disease	1,412 (3.0)	510 (2.0)	902 (4.0)	< 0.001
Immunocompromised patients	2,578 (5.4)	848 (3.4)	1,730 (7.6)	< 0.001

Data are presented as median and interquartile range or number (%), unless otherwise indicated

SARI Severe acute respiratory infection

[†]The capital region consists of Seoul, Incheon and Gyeonggi-do

Table 2 Treatment and outcomes of the patients

Variables	All patients	SARI pathogen detected	No SARI pathogen detected	*P*-value
Number of patients	47,857	25,132 (52.5)	22,725 (47.5)	
Treatment				
Antiviral agents	3,150 (6.6)	2,636 (10.5)	514 (2.3)	< 0.001
Antibiotics	28,417 (59.4)	13,065 (52.0)	15,352 (67.6)	< 0.001
Outcomes				
Pneumonia	24,168 (50.5)	12,231 (48.7)	11,937 (52.5)	< 0.001
Mortality	1,970 (4.1)	636 (2.5)	1,334 (5.9)	< 0.001
Length of hospital stay	5.0 (3.0–10.0)	4.0 (3.0–7.0)	7.0 (4.0–12.0)	< 0.001
ICU admission	3,037 (6.3)	1,385 (5.5)	1,652 (7.3)	< 0.001
Length of ICU stay	7.0 (3.0–14.0)	8.0 (4.0–16.0)	5.0 (3.0–12.0)	< 0.001

Data are presented as medians, interquartile ranges, and numbers (%), unless otherwise indicated

SARI Severe acute respiratory infection, *ICU* Intensive care unit

(2,576 cases, 5.4%), *Chlamydia pneumoniae* (173 cases, 0.4%), and *Legionella pneumophila* (87 cases, 0.2%).

Figure 1B–D shows the distribution of SARI pathogens by age. In children aged ≤ 7 years, the most prevalent pathogens were rhinovirus (22.8%), RSV (20.5%), and *Mycoplasma pneumoniae* (11.3%). School-aged children (8–18 years) had a high prevalence of *Mycoplasma pneumoniae* (23.1%), influenza (13.1%), and rhinovirus (12.2%). Among adults (19–64 years), SARS-CoV-2 (21.2%), *Streptococcus pneumoniae* (6.2%), and influenza (5.4%) were prevalent. In the older group (≥ 65 years), the most common pathogens were SARS-CoV-2 (13.0%),

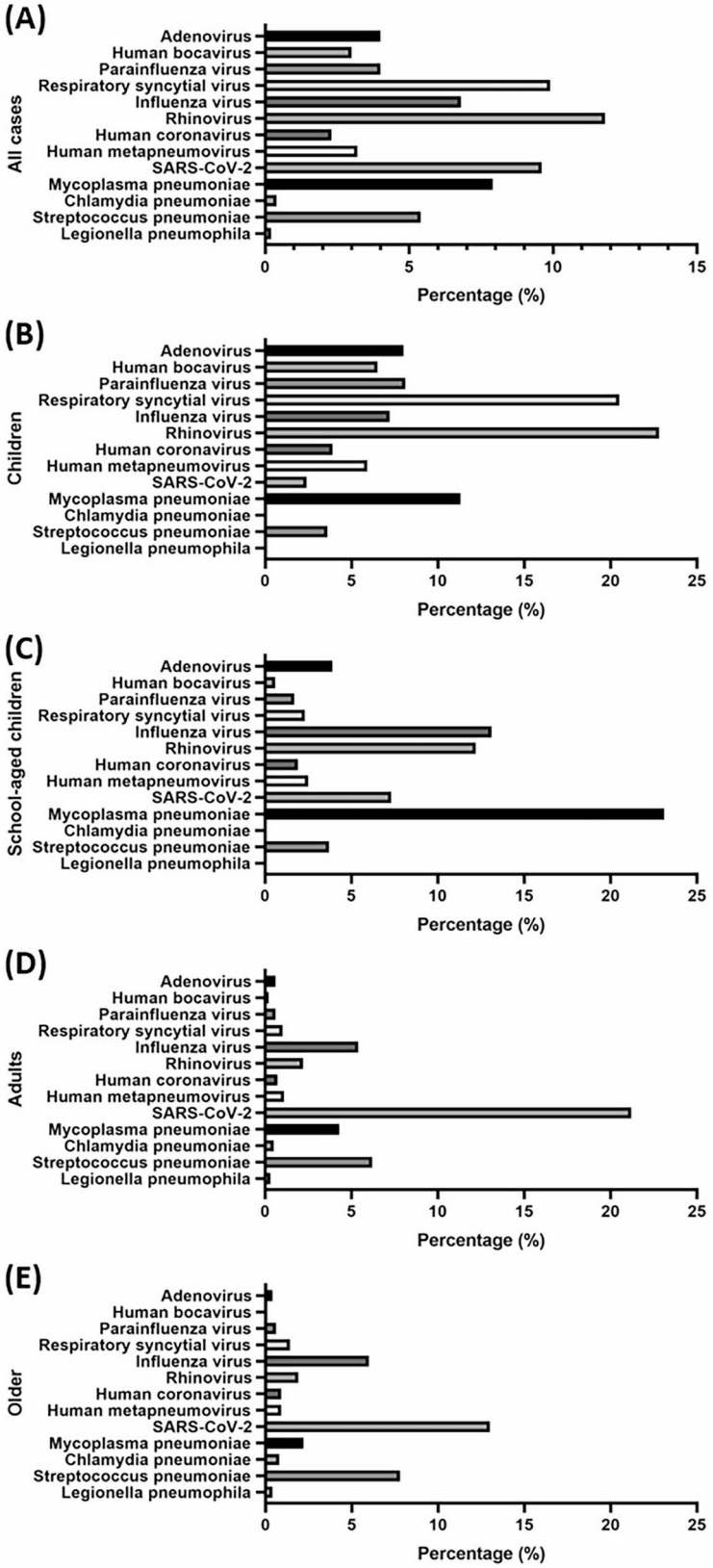


Fig. 1 Distribution of Respiratory Pathogens Identified in Patients with Severe Acute Respiratory Infection by Age Group. **A** Pathogen distribution among all cases. **B** Pathogen distribution among children (aged ≤7 years). **C** Pathogen distribution among school-aged children (aged 8-18 years). **D** Pathogen distribution among adults (aged 19–64 years). **E** Pathogen distribution in older adults (aged ≥ 65 years)

Streptococcus pneumoniae (7.8%), and influenza (6.0%) (Fig. 1E).

Furthermore, age-stratified analyses revealed distinct patterns across age groups. Antiviral use was consistently higher in pathogen-detected patients across all ages, while antibiotic use was more frequent in adults and older adults without detected pathogens. Pneumonia incidence was significantly higher in pathogen-detected patients across all age strata(all P <0.001). ICU admission and ICU length of stay were increased in pathogen-detected adults and older adults. Mortality was lower only in pathogen-detected adults (P <0.001), with no significant differences in other age groups. Hospital length of stay varied across age strata (Table S2).

SARI pathogen distribution by sex

Table 3 presents the distribution of SARI pathogens by sex. Among the participants, 57.6% (27,553) were male and 42.4% (20,304) were female. For viral pathogens, RSV, influenza, and human metapneumovirus had a significantly higher prevalence in females ($P<0.001$) than in male individuals. SARS-COV-2 also had a higher incidence in females than in male individuals (10.1% vs. 9.3%, $P=0.004$). Conversely, human coronavirus was more common in male than in female individuals ($P=0.003$). For bacterial pathogens, *Mycoplasma pneumoniae* was significantly more common in females than in male individuals (9.4% vs. 6.8%, $P<0.001$). *Chlamydia pneumoniae*, *Streptococcus pneumoniae*, and *Legionella pneumophila* were more common in male than in female individuals . These findings suggest sex differences in the distribution of certain SARI pathogens. In addition, we observed a significant age difference between male and female patients (median [IQR], 28.0 [2.0–71.0] vs. 12.0 [2.0–65.0]; $p<0.001$), which may partly explain the sex-related variation in pathogen distribution.

Geographic variation in SARI incidence by pathogen type

The Capital region comprises Seoul, Incheon, and Gyeonggi-do and has ~26 million residents (≈50% of South Korea's population). Of the 42 sentinel hospitals in our network, 23 are in the Capital region (Seoul $n=15$, Incheon $n=2$, Gyeonggi-do $n=6$) and 19 are in other regions (Table S1). Table 4 demonstrates the geographic variation in SARI incidence by pathogen type. Among the participants, 56.9% (27,223) were from the capital region, whereas 43.1% (20,634) were from non-capital regions. In the capital region, viral pathogens, such as RSV (10.6% vs. 8.9%, $P<0.001$), influenza (7.5% vs. 5.9%, $P<0.001$), human metapneumovirus (3.5% vs. 2.8%, $P<0.001$), and human bocavirus (3.3% vs. 2.6%, $P<0.001$), were more prevalent, whereas rhinovirus (11.0% vs. 12.8%, $P<0.001$), SARS-CoV-2 (8.4% vs. 11.2%, $P<0.001$), and parainfluenza virus (3.6% vs. 4.5%, $P<0.001$) were more common in non-capital regions. For bacterial pathogens, *Mycoplasma pneumoniae* was more frequent in capital regions, whereas *Streptococcus pneumoniae* and *Chlamydia pneumoniae* were more predominant in non-capital regions.

Comparison of SARI pathogen type on pneumonia, ICU admission, and mortality rates

Table S3 presents a distribution of SARI pathogens based on pneumonia occurrence. A comparative analysis of the

Table 3 SARI pathogen distribution by sex

Type	All	Male	Female	*P*-value
Number of patients	47,857	27,553 (57.6)	20,304 (42.4)	
Age (years)	18.0 (2.0–69.0)	28.0 (2.0–71.0)	12.0 (2.0–65.0)	< 0.001
SARI pathogen detected patients	25,132 (52.5)	14,006 (50.8)	11,126 (54.8)	< 0.001
Virus				
Adenovirus	1,922 (4.0)	1,125 (4.1)	797 (3.9)	0.385
Human bocavirus	1,430 (3.0)	791 (2.9)	639 (3.1)	0.079
Parainfluenza virus	1,903 (4.0)	1,065 (3.9)	838 (4.1)	0.147
Respiratory syncytial virus	4,723 (9.9)	2,546 (9.2)	2,177 (10.7)	< 0.001
Influenza virus	3,252 (6.8)	1,776 (6.4)	1,476 (7.3)	< 0.001
Rhinovirus	5,653 (11.8)	3,244 (11.8)	2,409 (11.9)	0.761
Human coronavirus	1,078 (2.3)	668 (2.4)	410 (2.0)	0.003
Human metapneumovirus	1,545 (3.2)	793 (2.9)	752 (3.7)	< 0.001
SARS-CoV-2	4,606 (9.6)	2,559 (9.3)	2,047 (10.1)	0.004
Bacteria				
Mycoplasma pneumoniae	3,765 (7.9)	1,861 (6.8)	1,904 (9.4)	< 0.001
Chlamydia pneumoniae	173 (0.4)	114 (0.4)	59 (0.3)	0.026
Streptococcus pneumoniae	2,576 (5.4)	1,581 (5.7)	995 (4.9)	< 0.001
Legionella pneumophila	87 (0.2)	71 (0.3)	16 (0.1)	< 0.001

Data are presented as medians, interquartile ranges, and numbers (%)

Table 4 Geographic variation in SARI incidence by pathogen type

Type	All	Capital region[†]	Non-capital region	*P*-value
Number of patients	47,857	27,223 (56.9)	20,634 (43.1)	
SARI pathogen detected patients	25,132 (52.5)	14,525 (53.4)	10,607 (51.4)	< 0.001
Virus				
Adenovirus	1,922 (4.0)	1,115 (4.1)	807 (3.9)	0.308
Human bocavirus	1,430 (3.0)	885 (3.3)	545 (2.6)	< 0.001
Parainfluenza virus	1,903 (4.0)	967 (3.6)	936 (4.5)	< 0.001
Respiratory syncytial virus	4,723 (9.9)	2,888 (10.6)	1,835 (8.9)	< 0.001
Influenza virus	3,252 (6.8)	2,041 (7.5)	1,211 (5.9)	< 0.001
Rhinovirus	5,653 (11.8)	3,008 (11.0)	2,645 (12.8)	< 0.001
Human coronavirus	1,078 (2.3)	631 (2.3)	447 (2.2)	0.268
Human metapneumovirus	1,545 (3.2)	965 (3.5)	580 (2.8)	< 0.001
SARS-CoV-2	4,606 (9.6)	2,295 (8.4)	2,311 (11.2)	< 0.001
Bacteria				
Mycoplasma pneumoniae	3,765 (7.9)	2,819 (10.4)	946 (4.6)	< 0.001
Chlamydia pneumoniae	173 (0.4)	37 (0.1)	136 (0.7)	< 0.001
Streptococcus pneumoniae	2,576 (5.4)	1,394 (5.1)	1,182 (5.7)	0.004
Legionella pneumophila	87 (0.2)	45 (0.2)	42 (0.2)	0.331

Data are presented as number (%)

[†]The capital region consists of Seoul, Incheon and Gyeonggi-do

prevalence of viral and bacterial pathogens in patients with pneumonia revealed that, except for SARS-CoV-2, most viral pathogens were less common, whereas bacterial pathogens were more prevalent.

Table S4 shows the distribution of SARI pathogens based on patients' admission to the ICUs. Among the various viral pathogens, SARS-CoV-2 was associated with a higher rate of ICU admissions, whereas other viruses, with the exception of influenza, were less commonly associated with ICU admission. In contrast, bacterial pathogens generally showed higher ICU admission rates, except for *Mycoplasma*, which demonstrated a lower rate of association with ICU admission.

Table S5 presents the distribution of SARI pathogens regarding patient mortality. Among viral pathogens, SARS-CoV-2 was associated with a higher mortality rate, whereas other viruses were less commonly associated with mortality. Among bacterial pathogens, *Mycoplasma pneumoniae* showed a lower mortality rate, whereas *Chlamydia pneumoniae*, *Streptococcus pneumoniae*, and *Legionella pneumophila* were associated with higher mortality rates.

Monthly pathogen distribution trends

Figure 2 and Table S6 indicate significant seasonal variations in SARI pathogens. Influenza and RSV peaked in winter (December–January), whereas rhinovirus circulated throughout the year with peaks in September–November. SARS-COV-2 showed fluctuating peaks, peaking in March and August. *Mycoplasma pneumoniae* exhibited periodic peaks during October–November, whereas *Streptococcus pneumoniae* showed little seasonality. Parainfluenza, bocavirus, and metapneumovirus showed minimal seasonal variation. *Chlamydia pneumoniae* showed no significant seasonality ($P = 0.345$).

Discussion

In this study, we used a nationwide surveillance dataset to examine the distribution of viral and bacterial pathogens associated with SARIs across demographic and geographic groups in South Korea. Data from more than 47,000 patients from 2017 to 2023 were analyzed to identify distinct pathogen profiles and clinical outcomes. The analysis revealed significant differences in pathogen prevalence based on age, region, and sex, as well as notable differences in clinical outcomes—including ICU admissions and mortality rates—associated with specific pathogens.

Analysis of age-related differences in SARI pathogen distribution identified distinct patterns across age groups. Among children, rhinovirus and RSV were the most frequently detected respiratory infection pathogens, consistent with the findings of previous studies [10]. Children exhibited the highest prevalence of *Streptococcus pneumoniae*, rhinovirus, and RSV infections, with the infant group (1–3 years) demonstrating significantly higher rhinovirus infection rates than those observed in other age groups. These findings are consistent with those of Kumar et al. [11], who reported rhinovirus (42%) and RSV (20%) as predominant in infants. *Mycoplasma pneumoniae* and influenza viruses were more common in school-aged children, consistent with findings by Zhang et al. [12], who identified influenza and adenoviruses as common pathogens of pediatric respiratory infections. Additionally, Kutty et al. [13] reported *Mycoplasma pneumoniae* in 8% of children (< 18 years) hospitalized

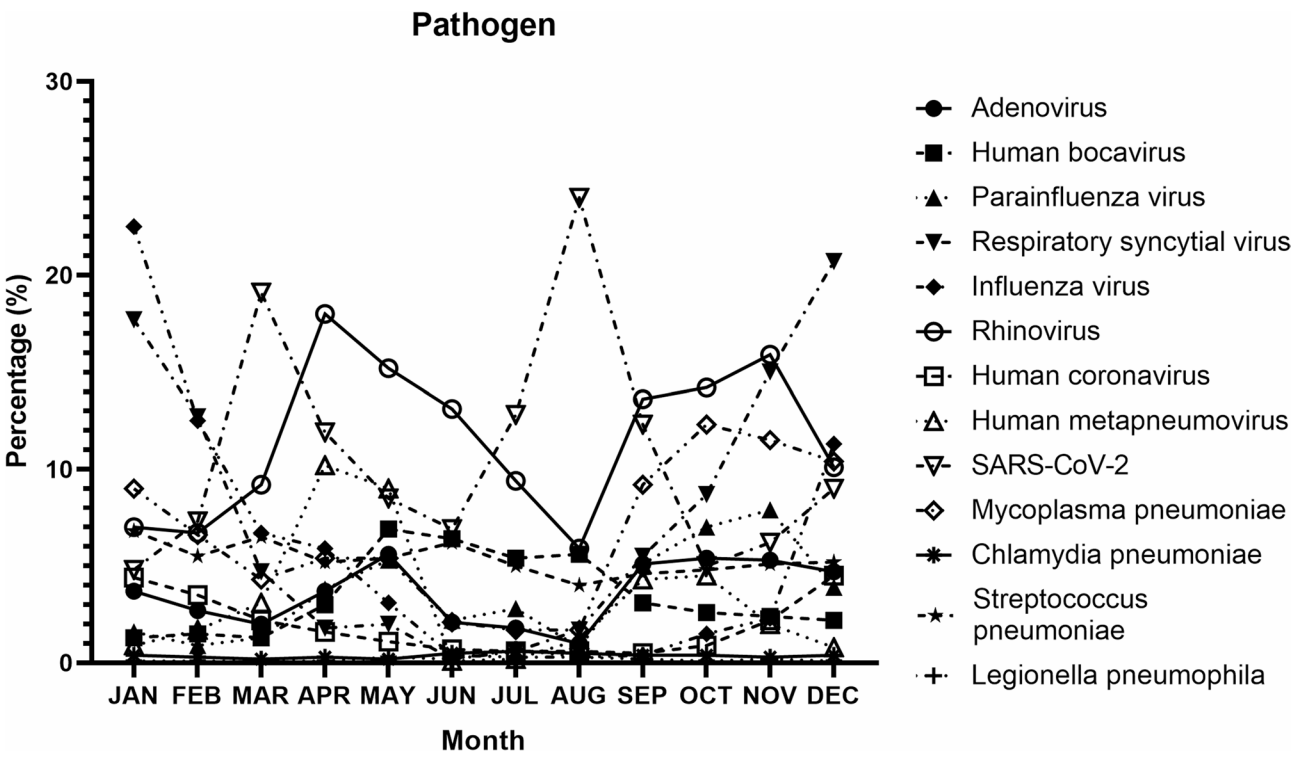


Fig. 2 Monthly distribution of severe acute respiratory infection pathogens

with community-acquired pneumonia, highlighting its role as an atypical bacterial pathogen. Similarly, Lv et al. [10] found that preschoolers and school-aged children had significantly higher infection rates for influenza A virus, *Chlamydia pneumoniae*, *Mycoplasma pneumoniae*, and *Haemophilus influenzae* compared with the rates observed in infants and toddlers, demonstrating age differences in pathogen distribution. In adults, SARS-CoV-2 and *Streptococcus pneumoniae* were predominant pathogens. Previous studies [14–16] in various regions have shown that respiratory infections caused by pathogens, such as influenza, rhinovirus, RSV, and *Mycoplasma pneumonia*, are common in adults. Additionally, SARS-COV-2 has shown high incidence rates among adults worldwide since its occurrence in 2019 [17]. In older patients, the most common pathogens were SARS-CoV-2, *Streptococcus pneumoniae*, and influenza. Other studies [4, 18] have identified pathogens, such as influenza virus, human rhinovirus, human parainfluenza virus, and *Streptococcus pneumonia*, as common causes of respiratory infections in older patients. These age-specific differences in pathogen prevalence demonstrate the importance of considering patient age in evaluating and managing respiratory infections.

The incidence of respiratory pathogens varies by sex and is potentially influenced by biological and behavioral factors [19, 20]. Behavioral factors, such as higher smoking and alcohol consumption rates among male individuals [21, 22], may increase related comorbidities and susceptibility to infections. Conversely, the higher participation of female individuals in childcare in Korea [23] may lead to greater exposure to outbreaks of pathogens associated with children. Additionally, intrinsic factors, including genetic and hormonal differences, may influence immune regulation and contribute to sex differences [24, 25] in pathogen prevalence.

Geographically, notable differences in pathogen prevalence were observed between capital and non-capital regions. Viral pathogens, such as RSV, influenza, human metapneumovirus, and human bocavirus, were more common in capital regions, whereas rhinoviruses, SARS-CoV-2, and parainfluenza viruses were more common in non-capital regions. Bacterial pathogens, such as *Mycoplasma pneumonia*, were more common in capital regions, whereas *Chlamydia pneumoniae* and *Streptococcus pneumoniae* were more common in non-capital regions. These geographic differences likely result from several factors influencing pathogen transmission. Improved access to health care in capital areas [26] may contribute to earlier detection, more effective treatment, and higher testing rates for certain infectious diseases.

Additionally, the higher population density of capital areas [27, 28] may result in more interpersonal contact, potentially increasing the spread of viral pathogens, such as RSV and influenza, which thrive in crowded environments. By contrast, in non-capital areas with a higher

proportion of older adults, the exposure risk may differ, especially for pathogens, such as pneumococci, which are more prevalent among older populations. Economic factors [29, 30] are also crucial, as individuals in capital areas with more economic resources and infrastructure often have better access to health care, leading to increased pathogen detection.

Our analysis revealed distinct differences in pathogen distribution associated with pneumonia occurrence, ICU admission, and mortality. Non-pneumonia cases were predominantly linked to pathogens, such as rhinovirus, RSV, and adenovirus, whereas pneumonia cases had a higher prevalence of *Streptococcus pneumoniae*, *Mycoplasma pneumoniae*, and SARS-CoV-2. Mortality and ICU admissions were more commonly associated with *Streptococcus pneumoniae*, *Legionella*, and SARS-CoV-2, indicating their role in severe diseases. Different viral pathogens are associated with distinct clinical outcomes, which are influenced by different factors, such as the specific viral agent, host immune response, and the presence of comorbidities. For example, in critically ill adults, viral pathogens are detected in up to 48.9% of severe respiratory tract infections but typically result in less severe outcomes compared with the outcome in bacterial infections [31].

In pediatric populations, RSV and human metapneumovirus are significant contributors to severe cases, whereas adenovirus is consistently associated with less clinical severity [32]. Among viral pathogens, SARS-CoV-2 was associated with significantly higher rates of ICU admission and patient mortality in our study. Since the onset of the COVID-19 pandemic, a high prevalence of SARS-CoV-2 infections in adults and older populations has been observed [33], aligning with findings that advanced age is a critical risk factor for severe disease progression and mortality in patients with SARS-COV-2 [34–36]. Older adults are particularly susceptible to infection owing to immune senescence [37] and a high prevalence of comorbidities [38, 39], which exacerbate the severity of respiratory infections and increase the likelihood of poor outcomes.

Among bacterial pathogens, *Streptococcus pneumoniae* and *Legionella* showed relatively high rates of ICU admission and mortality, in line with the findings of other studies [40–42]. This may be attributed to the fact that these bacteria can cause serious lower respiratory tract infections [43, 44], especially in immunocompromised patients or those with underlying medical conditions.

By contrast, *Mycoplasma pneumoniae* was common in the study group, but the incidence of serious outcomes was relatively low, likely owing to its prevalence in younger patients [45] with better prognoses [46]. These findings highlight the variability in disease progression and outcomes across different pathogens, emphasizing the need to consider pathogen-specific impact on clinical course and prognosis in the management of respiratory infections. Additionally, they offer a unique and large-scale perspective on the epidemiology of SARI pathogens in Korea, with implications for public health planning and respiratory infection control.

The observed seasonal variations in SARI pathogens provide critical insights for developing targeted public health preparedness and intervention strategies. As shown in Fig. 2, our results reveal distinct seasonal patterns: influenza and RSV showed a clear winter seasonality, whereas rhinovirus showed year-round circulation with pronounced peaks in early fall. These patterns are consistent with previous epidemiologic research on respiratory infections, and the winter seasonality of influenza and RSV is consistent with established research [47, 48], attributing these patterns to environmental factors, such as decreased temperatures, decreased humidity, and increased indoor crowding [49, 50]. Similarly, the year-round circulation of rhinovirus, with distinct peaks in the fall, reflects its remarkable adaptability to different environmental conditions [48, 51]. SARS-CoV-2 exhibited fluctuating transmission spikes, suggesting potentially complex interactions with evolving public health interventions and population immunity dynamics [52]. These temporal variations underscore the need for strategic approaches to respiratory disease management. Specifically, our findings highlight the importance of implementing seasonal vaccination programs [53], enhancing hospital preparedness during peak transmission periods, and developing targeted early diagnostic protocols for high-risk periods.

This study had several limitations. First, the analysis was limited to data from surveillance hospitals, which may have not accurately reflected pathogen prevalence in remote areas or populations with limited access to health care. Second, although all enrolled patients underwent pathogen testing, testing strategies varied by hospital and clinical presentation. Viral pathogens were primarily identified by PCR-based assays, whereas bacterial pathogens were detected by culture and urinary antigen testing. *Legionella* detection relied on urinary antigen testing or PCR-based assays rather than culture. Not all patients were tested for each pathogen, which might have contributed to variations in detection rates. Third, the number of participating hospitals increased during the study period, which could have led to variations in data collection and reporting practices. Differences in hospital-level diagnostic protocols and patient populations should be considered when interpreting trends over time. Fourth, the COVID-19 pandemic likely influenced surveillance patterns, with increased testing for respiratory infections potentially leading to higher pathogen detection rates. However, changes in hospital admission policies and

diagnostic priorities might have introduced variability in case reporting.

Overall, this study provides an in-depth analysis of SARI pathogens in South Korea, identifying key pathogen patterns, demographic influences, and clinical outcomes. These findings provide a basis for targeted prevention and treatment strategies, inform resource allocation, and highlight the continued need for robust pathogen surveillance to address the evolving nature of respiratory infections.

Conclusions

The study identified distinct seasonal and temporal trends in severe acute respiratory infection (SARI) pathogens in South Korea, with notable shifts in the prevalence of viruses and bacteria over time. Demographic and clinical influences: Age, sex, and comorbidities significantly influenced infection rates and clinical outcomes, with vulnerable groups such as young children and the elderly experiencing higher severity and hospitalization rates. These findings support the development of targeted prevention and treatment strategies, optimized allocation of healthcare resources, and highlight the critical role of continuous pathogen surveillance in managing evolving respiratory threats.

Abbreviations

COVID-19	Coronavirus disease 2019
RSV	Respiratory syncytial virus
SARIs	Severe acute respiratory infections
KDCA	Korean Disease Control and Prevention Agency
BALF	Bronchoalveolar lavage fluid
PCR	Polymerase chain reaction
ICU	Intensive care unit

Supplementary Information

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Supplementary Material 1.

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Not applicable.

Authors' contributions

All authors had full access to all the study data and take responsibility for the integrity of the data and the accuracy of the data analysis. All authors have read and approved the final manuscript. Concept and design: DHK, SYK, SJL, SIL Acquisition, analysis, or interpretation of data: All authors. Drafting of the manuscript: DHK, SJL, SIL. Critical review of the manuscript for important intellectual content: DHK, SIL. Statistical analysis: DHK, SJL, SIL. Obtained funding: YSL, JYM, SIL. Administrative, technical, or material support: KDK, HRC, JER. Supervision: SJL, SIL.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the ethical standards of the Declaration of Helsinki. The Institutional Review Board of Chungnam National University Hospital (No. 2023-03-076) reviewed and approved the study and waived the requirement of informed consent. All patient data were anonymized and treated with strict confidentiality to protect the privacy of the participants. Identifiable information was removed before analysis to ensure compliance with privacy regulations.

Competing interests

The authors declare no competing interests.

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